

DSC540WeekProjectMilestone2

April 24, 2025

1 Project Milestone 2

```
[7]: # Importing Pandas packages required for this exercise.
```

```
import pandas as pd
import seaborn as sns
import matplotlib

import matplotlib.dates as mdates
from matplotlib.dates import DateFormatter
import matplotlib.ticker as tick
```

```
[28]: # loading covid CSV file for data transformation and cleaning
```

```
covid = pd.read_csv("./time_series_covid_19_confirmed.csv")
covid.head()
```

```
[28]: Province/State Country/Region    Lat    Long  1/22/20  1/23/20  1/24/20  \
0      NaN      Afghanistan  33.0000  65.0000      0      0      0
1      NaN      Albania    41.1533  20.1683      0      0      0
2      NaN      Algeria    28.0339   1.6596      0      0      0
3      NaN      Andorra    42.5063   1.5218      0      0      0
4      NaN      Angola    -11.2027  17.8739      0      0      0

      1/25/20  1/26/20  1/27/20  ...  4/4/20  4/5/20  4/6/20  4/7/20  4/8/20  \
0      0      0      0  ...    299    349    367    423    444
1      0      0      0  ...    333    361    377    383    400
2      0      0      0  ...   1251   1320   1423   1468   1572
3      0      0      0  ...    466    501    525    545    564
4      0      0      0  ...     10     14     16     17     19

      4/9/20  4/10/20  4/11/20  4/12/20  4/13/20
0      484      521      555      607      665
1      409      416      433      446      467
2     1666     1761     1825     1914     1983
3      583      601      601      638      646
4       19       19       19       19       19
```

[5 rows x 87 columns]

```
[30]: # Dropping lat.lon columns not required in this project from file
```

```
coviddrop = covid.drop(['Lat','Long'],axis = 1)
coviddrop
```

```
[30]:
```

	Province/State	Country/Region	1/22/20	1/23/20	\
0	NaN	Afghanistan	0	0	
1	NaN	Albania	0	0	
2	NaN	Algeria	0	0	
3	NaN	Andorra	0	0	
4	NaN	Angola	0	0	
..	...	...	...	...	
259	Saint Pierre and Miquelon	France	0	0	
260	NaN	South Sudan	0	0	
261	NaN	Western Sahara	0	0	
262	NaN	Sao Tome and Principe	0	0	
263	NaN	Yemen	0	0	

	1/24/20	1/25/20	1/26/20	1/27/20	1/28/20	1/29/20	...	4/4/20	\
0	0	0	0	0	0	0	...	299	
1	0	0	0	0	0	0	...	333	
2	0	0	0	0	0	0	...	1251	
3	0	0	0	0	0	0	...	466	
4	0	0	0	0	0	0	...	10	
..	...	...	...	...	...	...	...	...	
259	0	0	0	0	0	0	...	0	
260	0	0	0	0	0	0	...	0	
261	0	0	0	0	0	0	...	0	
262	0	0	0	0	0	0	...	0	
263	0	0	0	0	0	0	...	0	

	4/5/20	4/6/20	4/7/20	4/8/20	4/9/20	4/10/20	4/11/20	4/12/20	\
0	349	367	423	444	484	521	555	607	
1	361	377	383	400	409	416	433	446	
2	1320	1423	1468	1572	1666	1761	1825	1914	
3	501	525	545	564	583	601	601	638	
4	14	16	17	19	19	19	19	19	
..	...	...	...	...	...	...	...	...	
259	1	1	1	1	1	1	1	1	
260	1	1	2	2	3	4	4	4	
261	4	4	4	4	4	4	4	6	
262	0	4	4	4	4	4	4	4	
263	0	0	0	0	0	1	1	1	

```

4/13/20
0      665
1      467
2     1983
3      646
4       19
..      ...
259     1
260     4
261     6
262     4
263     1

```

[264 rows x 85 columns]

[32]: *# Selecting 4 countries that has highest number of cases*

```

covidMelt = coviddrop.melt(id_vars = ['Country/Region'])
coviddropNan = covidMelt.dropna()
coviddropvariable = coviddropNan[coviddropNan['variable'] != 'Province/State']
covidtopCountries = coviddropvariable.groupby(['Country/Region'])['value'].
    ↪sum().reset_index().sort_values(by = 'value',ascending = False)
covidtopCountries[0:5]

```

```

[32]:   Country/Region   value
171          US  6278243
36         China  5262621
84         Italy  2977220
156        Spain  2574182
65        Germany  1986317

```

```

[17]: Covidtop4countries = coviddropvariable.loc[coviddropvariable['Country/Region'].
    ↪isin(['US','Brazil','Russia','Spain'])]
set(Covidtop4countries['Country/Region'])

```

[17]: {'Brazil', 'Russia', 'Spain', 'US'}

[34]: *# Renaming country date columns and Parsing date and grouping data*

```

import numpy as np
pd.set_option('mode.chained_assignment',None)
Covidtop4countries.columns = ['Country','Date','No of Cases']
Covidtop4countries['Date'] = pd.to_datetime(Covidtop4countries.Date)
Covidtop4countriesGroup = Covidtop4countries.groupby(['Country',pd.Grouper(key_
    ↪= 'Date', freq = 'SM'))]['No of Cases'].sum().reset_index().
    ↪sort_values('Date')
Covidtop4countriesGroup

```

```

Covidtop4countriesGroup1 =
    Covidtop4countriesGroup[Covidtop4countriesGroup['Date'] < '2020-06-30']
Covidtop4countriesGroup1
Covidtop4countriesGroup1['Parsed Date'] = Covidtop4countriesGroup1['Date'].
    astype(str)
Covidtop4countriesGroup1['Parsed Date'] = Covidtop4countriesGroup1['Parsed_
    Date'].str[6:]
Covidtop4countriesGroup1['Parsed Date'] = Covidtop4countriesGroup1['Parsed_
    Date'].replace({'-':'/'}, regex=True)
Covidtop4countriesGroup1[0:10]

```

```

/var/folders/18/ygzmjv90m72g8_1d1sfk1b80000gn/T/ipykernel_43399/2231736262.py:7
: FutureWarning: 'SM' is deprecated and will be removed in a future version,
please use 'SME' instead.

```

```

Covidtop4countriesGroup = Covidtop4countries.groupby(['Country',pd.Grouper(key
= 'Date', freq = 'SM'))['No of Cases'].sum().reset_index().sort_values('Date')

```

```

[34]:
Country      Date No of Cases Parsed Date
0  Brazil 2020-01-15      0      1/15
12 Spain 2020-01-15      0      1/15
18  US 2020-01-15     31      1/15
6  Russia 2020-01-15      0      1/15
13 Spain 2020-01-31     20      1/31
7  Russia 2020-01-31     30      1/31
19  US 2020-01-31    161      1/31
1  Brazil 2020-01-31      0      1/31
8  Russia 2020-02-15     28      2/15
2  Brazil 2020-02-15      3      2/15

```

```

[36]: # Applying stack function and transforming the columns accordingly.

```

```

covidUS = covid[covid['Country/Region'] == 'US']
covidLatLong = covidUS.drop(['Lat','Long'],axis = 1)
Covidtransform = covidLatLong.stack()
CovidtransformF = Covidtransform.reset_index()
CovidtransformF

```

```

[36]:
level_0      level_1      0
0      225 Country/Region  US
1      225      1/22/20      1
2      225      1/23/20      1
3      225      1/24/20      2
4      225      1/25/20      2
..      ...      ...
79     225      4/9/20  461437
80     225      4/10/20  496535
81     225      4/11/20  526396

```

```
82      225      4/12/20  555313
83      225      4/13/20  580619
```

[84 rows x 3 columns]

```
[23]: # renaming columns and sorting the data afterwards
CovidtranformF = CovidtranformF[CovidtranformF['level_1'] != 'Country/Region']
CovidtranformFilter = CovidtranformF[['level_1',0]]
CovidtranformFilter.columns = ['Date','No of Cases']
CovidtranformFilter = CovidtranformFilter.sort_values(by = 'No of_
↳Cases',ascending = False)
CovidtranformFilter
```

```
[23]:      Date No of Cases
83  4/13/20      580619
82  4/12/20      555313
81  4/11/20      526396
80  4/10/20      496535
79  4/9/20       461437
..      ...      ...
6   1/27/20         5
3   1/24/20         2
4   1/25/20         2
2   1/23/20         1
1   1/22/20         1
```

[83 rows x 2 columns]

```
[38]: Covidsort = CovidtranformFilter.sort_values(by = 'No of Cases',ascending =_
↳False)
Covidsort
```

```
[38]:      Date No of Cases
83  4/13/20      580619
82  4/12/20      555313
81  4/11/20      526396
80  4/10/20      496535
79  4/9/20       461437
..      ...      ...
6   1/27/20         5
3   1/24/20         2
4   1/25/20         2
2   1/23/20         1
1   1/22/20         1
```

[83 rows x 2 columns]

[]: